

GATE IN

Instrumentation Engineering

TREND-BASED MOCK PAPER

— 02 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	8 subjects

Difficulty Distribution

Easy: 23 questions (35%)

Moderate: 30 questions (46%)

Difficult: 12 questions (18%)

GENERAL INSTRUCTIONS

GATE IN — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
-

3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
-

4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
-

5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
-

Q1. 1M [-0.33]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q2. 1M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

Enter your answer here

Q3. 1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q4. 1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q5. 1M [-0.33]

[Analog & Digital Electronics]

8-bit ADC has resolution:

- (A) $1/256$
- (B) $1/128$
- (C) $1/512$
- (D) $1/64$

Q6. 1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q7.

1M [-0.33]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q8.

1M [-0.33]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q9.

1M [no -ve]

[Measurement Systems]

Moving coil instrument measures:

*Enter your answer here***Q10.**

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q11.

1M [no -ve]

[General Aptitude]

20% of 200:

*Enter your answer here***Q12.**

1M [no -ve]

[Sensors & Instrumentation]

LVDT measures:

*Enter your answer here***Q13.**

1M [no -ve]

[General Aptitude]

20% of 200:

Enter your answer here

Q14.

1M [-0.33]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q15.

1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q16.

1M [-0.33]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q17.

1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q18.

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q19.

1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q20.

1M [no -ve]

[Communications & Process Control]

FM uses:

*Enter your answer here***Q21.**

1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q22.

1M [no -ve]

[Signals and Systems]

FFT computes:

*Enter your answer here***Q23.**

1M [no -ve]

[Measurement Systems]

Moving coil instrument measures:

*Enter your answer here***Q24.**

1M [-0.33]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q25.

1M [no -ve]

[Measurement Systems]

Moving coil instrument measures:

*Enter your answer here***Q26.**

1M [no -ve]

[Analog & Digital Electronics]

Instrumentation amp has:

*Enter your answer here***Q27.**

1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q28.

1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q29.

1M [-0.33]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q30.

1M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

*Enter your answer here***Q31.**

2M [-0.66]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q32.

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q33.

2M [-0.66]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q34.

2M [no -ve]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q35.

2M [no -ve]

[Control Systems]

PID controller has:

*Enter your answer here***Q36.**

2M [-0.66]

[Measurement Systems]

Moving coil instrument measures:

- (A) AC only
- (B) DC only
- (C) Both
- (D) None

Q37.

2M [-0.66]

[Measurement Systems]

Moving coil instrument measures:

- (A) AC only
- (B) DC only
- (C) Both
- (D) None

Q38.

2M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

*Enter your answer here***Q39.**

2M [-0.66]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q40.

2M [-0.66]

[Analog & Digital Electronics]

8-bit ADC has resolution:

- (A) $1/256$
- (B) $1/128$
- (C) $1/512$
- (D) $1/64$

Q41.

2M [-0.66]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q42.

2M [no -ve]

[Analog & Digital Electronics]

Instrumentation amp has:

Enter your answer here

Q43.

2M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

Enter your answer here

Q44.

2M [no -ve]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q45.

2M [no -ve]

[General Aptitude]

20% of 200:

Enter your answer here

Q46.

2M [-0.66]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q47.

2M [-0.66]

[Measurement Systems]

Moving coil instrument measures:

- (A) AC only
- (B) DC only
- (C) Both
- (D) None

Q48.

2M [-0.66]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q49.

2M [-0.66]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q50.

2M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q51.

2M [-0.66]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q52.

2M [-0.66]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q53.

2M [no -ve]

[Communications & Process Control]

Closed-loop uses:

Enter your answer here

Q54.

2M [-0.66]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q55.

2M [-0.66]

[Analog & Digital Electronics]

8-bit ADC has resolution:

- (A) $1/256$
- (B) $1/128$
- (C) $1/512$
- (D) $1/64$

Q56.

2M [-0.66]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q57.

2M [-0.66]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q58.

2M [-0.66]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q59.

2M [-0.66]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q60.

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q61.

2M [-0.66]

[Analog & Digital Electronics]

8-bit ADC has resolution:

- (A) $1/256$
- (B) $1/128$
- (C) $1/512$
- (D) $1/64$

Q62.

2M [no -ve]

[Analog & Digital Electronics]

Instrumentation amp has:

*Enter your answer here***Q63.**

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q64.

2M [no -ve]

[Measurement Systems]

Moving coil instrument measures:

*Enter your answer here***Q65.**

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None